

Car Head-Up Display Based SMART INFOTAINMENT SYSTEM



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Let us imagine a fine day when you are going with family to a picnic in a car. Music and movies keep the kids entertained. When they need a change of song or movie, they just wave through the air. Meanwhile, the family member driving the car can ensure that the car is moving on the right path and at the right speed, without taking eyes off the road, even to glance at the GPS.

Embitel's solution to enable a smarter and safer vehicle features such a smart infotainment system that visually integrates with the car's windshield. The system also connects to the vehicle's electronics and leverages the Android experience to which any user could easily get accustomed.

While discussing innovations at Embitel with Vidya Sagar, who heads connected car and automation-related initiatives, an interesting story of how innovative concepts could be engineered to suit business reality unfolded.

Let us take a look at the journey the innovators took.

Bringing life into the system

About six years ago, the engineers at Embitel found the need for an in-vehicle infotainment (IVI) system that would bundle the media requirements of drivers as well as the passengers in a car. At that time, most people were enthralled by the introduction of Android based smartphones. These smartphones housed operating functionalities that fulfilled all their media and connectivity

requirements with an interactive graphical user interface (GUI).

As the flame of ideas grew brighter, teams at Embitel realised that they must build their infotainment system around Android operating system due to its open source nature and easy integration with third-party stacks. Past expertise and in-depth analysis were taken into account and the project for creating an automotive multimedia application took shape.

In the long run, this decision proved to be a viable business case. Embitel's product engineering services team successfully delivered their first automotive infotainment system in the year 2014 for a tier 1 supplier.

What's surprising is that they were able to do this while Google themselves are yet to launch Android N for automotive systems, with just a prototype demonstration available as of May 2016.

The connectivity aspect

Data is acquired from onboard diagnostic board and controller area network bus on the vehicle. Every typical car has these on board, which is a network of points. There are electronic control units within the vehicle, which control doors, vipers, power train and so on. These provide crucial data to the hub, and the processing follows.

Added awareness. The product's awareness features work on data analytics, driven by two major paradigms. One focuses on real-time analytics for quick decision making and the other works on studying the pattern

of usage in order to benchmark against reported events or trend analysis. Once the trend analysis is inferred, predictive analytics can also be enforced.

On another track, companies providing maintenance, insur-

Fig. 1: Car HUD layout and design

